

ROBO RUMBLE

Autonomous 4WD Racing Robot

A compact, collision-aware platform built to sense, decide and drive without a human controller.

4

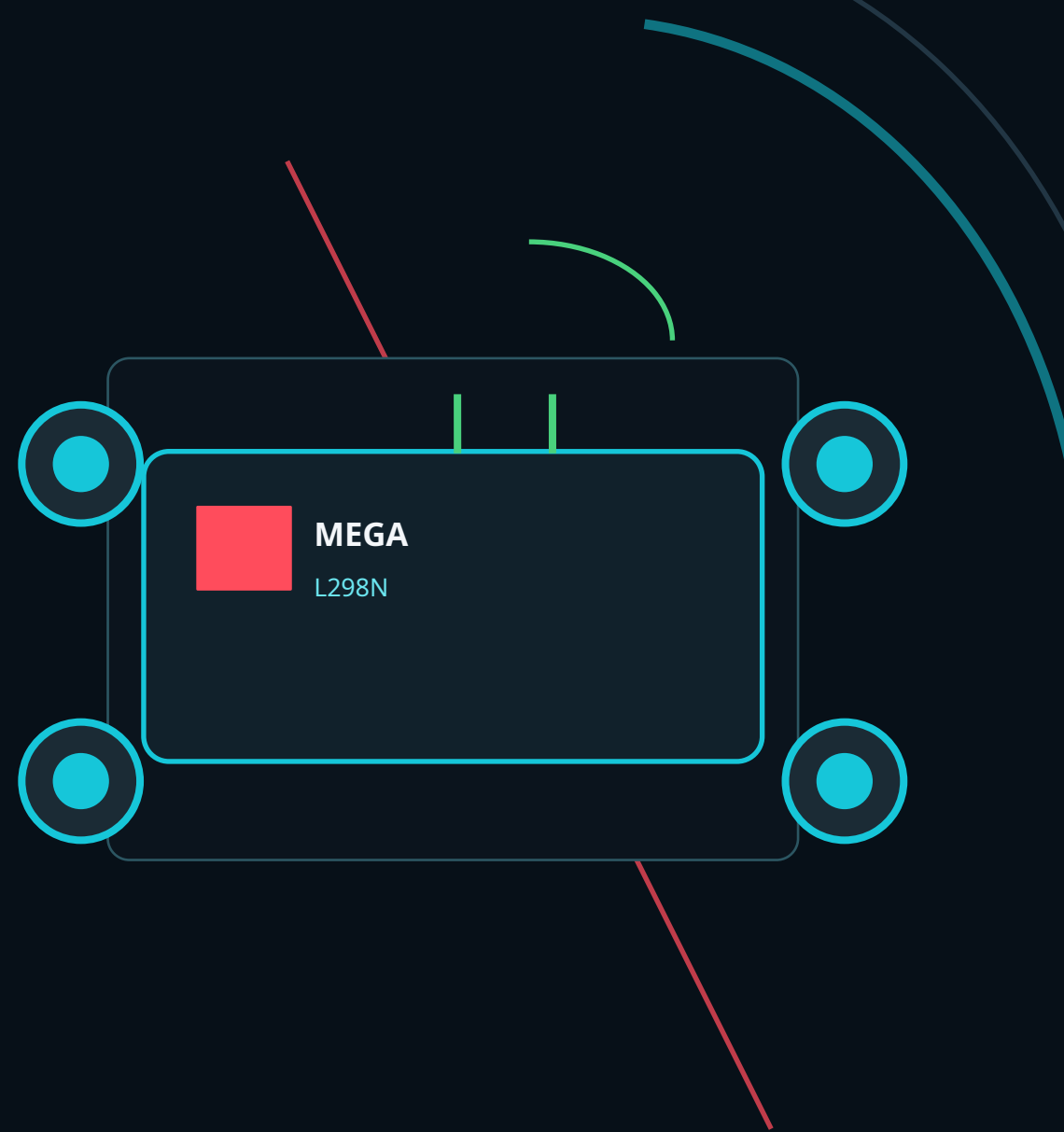
drive motors

3

ultrasonic zones

100%

autonomous



Autonomous racing needs more than speed

The challenge is maintaining momentum while reacting to changing track conditions in real time.



REACTION TIME

A manually controlled car depends on human reaction - too slow for sudden obstacles or tight turns.



LINE-OF-SIGHT

An RC operator sees only what the camera or eyes reveal. The robot cannot independently interpret its path.



CONTROL + POWER

Four-wheel traction creates current demand and control complexity. Poor integration causes stalls, drift and resets.

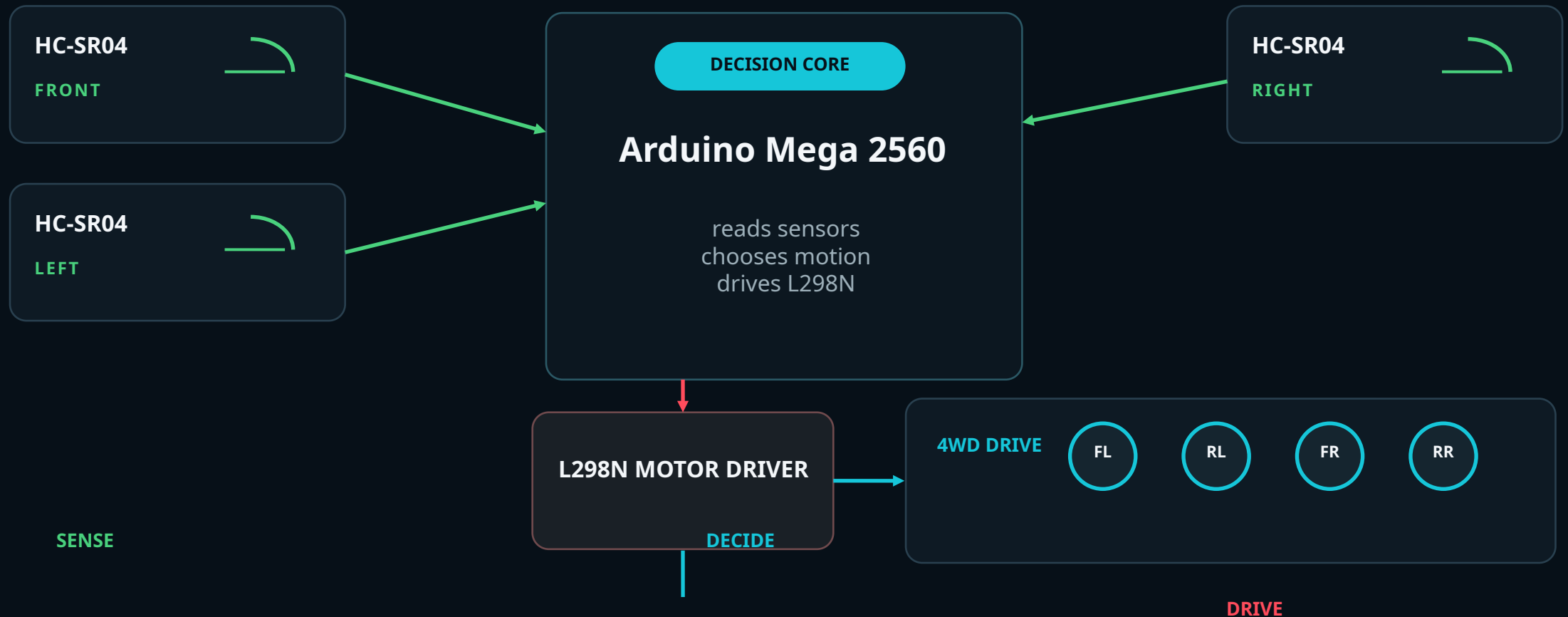
DESIGN TARGET

Sense the environment -> decide the safest racing action -> drive all four wheels -> keep going.

Competition brief used: compact footprint, battery-powered, fully autonomous operation.

A three-zone sensing system wrapped around a 4WD drive platform

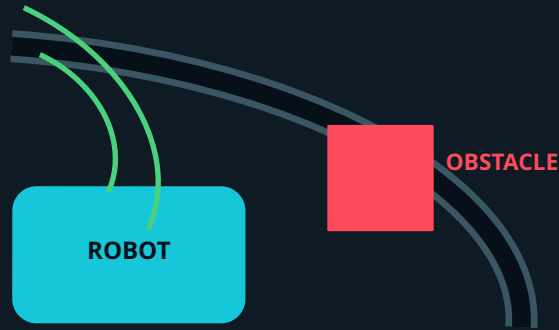
The robot continuously measures its surroundings and converts distance data into motor commands.



Architecture follows the supplied Fritzing concept: 3 x HC-SR04 + Arduino Mega 2560 + L298N + 4 motors.

Simple rules. Fast decisions. Fewer collisions.

The first version prioritises reliable obstacle-aware behaviour rather than fragile high-speed complexity.



- 1 Measure** Read front / left / right distance.
- 2 Classify** Clear path? blocked front? tight side?
- 3 Choose** Throttle straight or bias away from obstacle.
- 4 Correct** Re-check immediately after the manoeuvre.

Race goal: maximise completed laps / runs by reducing avoidable impact events and driver intervention.

A competition robot can become a reusable robotics platform

The same sensing-and-drive architecture transfers to education, demos and lightweight indoor automation.

STEM TRAINING

A practical platform for sensors, embedded programming, motor control and autonomous decision-making.

LEARN

ROBOTICS EVENTS

A visible, exciting demonstration of autonomous navigation that can be tuned for different tracks.

COMPETE

INDOOR MOBILITY

A starting point for simple obstacle-aware carts, inspection platforms or guided material movement.

DEPLOY

VALUE PROPOSITION Low-cost autonomy + modular electronics + reusable control logic = a platform worth improving beyond race day.

Commercial framing is a project proposition, not a claim of current commercial deployment.

Built with accessible off-the-shelf electronics

The current BOM stays within a student-project budget while keeping the platform serviceable and upgradeable.

TOTAL PROJECT BOM

R4,432.43

including VAT

Core BOM total from the current procurement sheet.

BUDGET CONTROL

COST DRIVERS

4 x drive motors

~R2,285



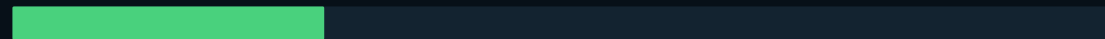
Mega + sensors + L298N

~R1,485



Power + wiring + mechanics

~R662



POWER NOTE: the 9V PP3 shown in the reference circuit is not suitable for sustained four-motor operation; use a higher-current rechargeable pack in the finished race robot.

Pricing basis: current BOM sheet, ZAR; verify supplier stock and final cart pricing before purchase.

The person behind the machine

Hands-on integration across mechanics, electronics, firmware and race testing.



Kgaogelo "KayGee"

TEAM LEAD

Project direction + race strategy

EMBEDDED SYSTEMS

Arduino control + sensor logic

MECH / ELECTRICAL

Drive train + wiring integration

TEST / TUNING

Track trials + decision-rule tuning

MISSION

Build an autonomous robot that is fast enough to compete, simple enough to repair, and smart enough to finish.

Team slide uses the uploaded member photo supplied in this conversation.